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DETAIL SPECIFICATION

HARDNESS TESTING REAGENTS, BOILER FEEDWATER, SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the composition and quality of chemical reagents used for testing hardness in shipboard boiler feedwater.

1.2 Classification. Hardness testing chemicals are of the following classes, as specified (see 6.2):

a. Class A – Hardness titrating solution

b. Class B – Hardness buffer solution

c. Class C – Hardness indicator powder

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-20135 - Sugar, White, Refined, and Sugar, Brown

DEPARTMENT OF DEFENSE SPECIFICATIONS

DOD-R-23679 - Reagents, Standard, Water Test Chemicals (Metric)

(Copies of these documents are available online at https://quicksearch.dla.mil.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS 411-1 - Hazardous Material Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN CHEMICAL SOCIETY (ACS)

ACS Reagent Chemicals: Specifications and Procedures for Reagents and Standard-Grade Reference Materials

(Copies of this document are available online at https://pubs.acs.org/.)

ASTM INTERNATIONAL

ASTM D1193 - Standard Specification for Reagent Water

ASTM D1293 - Standard Test Methods for pH of Water

ASTM E169 - Standard Practices for General Techniques of Ultraviolet-Visible Quantitative Analysis

ASTM E287 - Standard Specification for Laboratory Glass Graduated Burets

ASTM E288 - Standard Specification for Laboratory Glass Volumetric Flasks

ASTM E969 - Standard Specification for Glass Volumetric (Transfer) Pipets

ASTM E1252 - Standard Practice for General Techniques for Obtaining Infrared Spectra for Qualitative Analysis

(Copies of these documents are available online at www.astm.org.)

STANDARD METHODS

4500-S^{2-} - Sulfide

(Copies of this document are available online at www.standardmethods.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Reagent material quality. Materials used in the manufacture of each of the reagents covered by this specification shall conform to all the applicable specifications of the American Chemical Society (ACS). Reagent solutions shall be prepared using reagent water in accordance with ASTM D1193, type II.

3.1.1 Prohibited materials. The reagent chemicals shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS 411-1 for all other programs.

3.1.2 Class A, hardness titrating solution. The hardness titrating solution shall be a 20 equivalents per million (epm) solution of disodium ethylenediaminetetraacetate (Na_2EDTA) conforming to the characteristics in table I when tested as specified herein, with any required pH adjustment accomplished using sodium hydroxide or hydrochloric acid.

TABLE I. Class A, hardness titrating solution characteristics.

Characteristic	Minimum	Maximum	Test Paragraph
Concentration, epm	19.60	20.40	4.2.2.2
pH	8.8	10.0	4.2.2.3

3.1.3 Class B, hardness buffer solution. The hardness buffer shall be a clear liquid conforming to the characteristics in table II when tested as specified herein.

TABLE II. Class B, hardness buffer solution characteristics.

Characteristic	Minimum	Maximum	Test Paragraph
Performance, initial pH	10.0	11.0	4.2.3.1
Performance, final pH	10.0	11.0	4.2.3.1
Performance, epm	0.17	0.23	4.2.3.1

3.1.4 Class C, hardness indicator powder. The hardness indicator shall be a dry, water soluble, free-flowing powder, consisting of a uniform mixture of Eriochrome Black T (EBT) (Chemical Abstracts Service [CAS] Number 1787-61-7), or equal, and sucrose in accordance with A-A-20135, type I, style A and shall conform to the characteristics in table III when tested as specified herein.

TABLE III. Class C, hardness indicator powder characteristics.

Characteristic	Minimum	Maximum	Test Paragraph
EBT, or equal	0.12	0.30	4.2.1
Sucrose	Detected	—	4.2.4.2
Performance, epm	0.17	0.23	4.2.4.3
Appearance	Granular, free flowing	—	4.2.4.4

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4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall consist of the samples specified in table IV for the tests of 4.2.1 through 4.2.4.4.

Table IV. Required samples for Government acceptance testing.

Class	Description	Samples per Lot
Class A	Hardness titrating solution	Three 1-liter bottles
Class B	Hardness buffer solution	Three 120-milliliter bottles
Class C	Hardness indicator powder	Three 100-gram bottles

4.1.1 Prohibited materials. Prohibited materials shall be verified as specified (see 3.1.1 and 6.2).

4.1.2 Lot. A lot shall consist of all solutions or chemicals from one batch. In cases in which it is not practicable to define the sampling lot in this manner, the lot shall consist of all solutions or chemicals of one class for delivery at one time.

4.1.3 Sampling for tests. Sampling of a lot for test purposes shall be as specified in table IV.

4.1.4 Examination of filled containers. Each sample filled container selected as specified in 4.2.2 shall be inspected to verify compliance with all requirements of this specification. Containers shall be examined for defects of the container and closure, for evidence of leakage, for unsatisfactory markings, and for the required amount of reagent.

4.1.5 Rejection of lots. If a sample fails to pass any of the tests specified in table IV, the production lot represented by that sample shall be rejected.

4.2 Tests.

4.2.1 Test glassware, reagents, and conditions. Volumetric glassware used in the test procedures shall be class A glassware in accordance with ASTM E287, ASTM E288, or ASTM E969, as applicable. Water used in the test procedures shall be in accordance with ASTM D1193, type II reagent water. Test solutions shall be prepared from reagent grade chemicals in accordance with ACS Reagent Chemicals: Specifications and Procedures for Reagents and Standard-Grade Reference Materials. Titrations shall be conducted at 20 to 25 °C. Ultraviolet-visible quantitative analysis shall be conducted in accordance with ASTM E169, using either a double-beam or single-beam instrument. The use of a visible-spectrum (Vis) instrument is acceptable for these tests.

4.2.2 Class A, hardness titrating solution.

4.2.2.1 Preparation of test reagents.

4.2.2.1.1 Calcium standard. The preparation of the calcium standard shall be as follows:

a. Dry reagent grade calcium carbonate ($CaCO_3$) at 300 °C for 4 hours and cool in a desiccator.

b. To a 1-liter volumetric flask, add 5 milliliters (mL) of reagent water, 1 mL of concentrated hydrochloric acid, and then 0.2000 gram of $CaCO_3$.

c. Swirl to dissolve. Then dilute to the mark with reagent water.

4.2.2.1.2 Ammonium hydroxide (NH_4OH) solution. 40 mL of reagent water shall be mixed with 10 mL of reagent grade ammonium hydroxide.

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4.2.2.2 Concentration. The procedure for calculating the concentration of EDTA shall be as follows:

a. To a 1-liter beaker, add 600 mL of reagent water, 40 drops of NH_4OH solution, and 8 scoops of hardness indicator meeting class C of this specification.

b. Transfer 75 mL of this solution into each of five 250-mL Erlenmeyer flasks.

c. Select one flask to be a blank; dilute with reagent water to 100 mL., and then add an additional 2 mL of titrant meeting class A of this specification.

d. To the remainder of the flasks, add 25 mL of calcium standard described in 4.2.2.1.1.a.

e. Titrate using a 10-mL burette using the color standard as a comparison for the endpoint.

f. Record the volume of titrant used, and calculate concentration of EDTA in epm by dividing 99.91 by the volume of titrant.

4.2.2.3 pH. The pH of the hardness titrating solution shall be measured in accordance with ASTM D1293.

4.2.3 Class B, hardness buffer solution.

4.2.3.1 Hardness buffer performance. The test shall be performed using the hardness buffer solution being tested, hardness titrant meeting the requirements of class A of this specification, hardness indicator meeting class C of this specification, and hardness standard solution in accordance with class E of DOD-R-23679. The procedure for calculating the hardness of the hardness buffer solution shall be as follows:

a. Measure 100 mL of the hardness standard solution using a volumetric pipet, and transfer into either a white porcelain casserole or into a clear Erlenmeyer flask placed on a white background.

b. Add 2 mL of buffer solution and 1 dipper of hardness indicator to a casserole.

c. Measure the beginning pH in accordance with the procedures in ASTM D1293. Titrate to the endpoint, and measure the pH final endpoint.

d. Calculate the hardness of the hardness standard using the following equation:

$$H_1 = \frac{V_2 N_2}{V_1}$$

Where:

H_1 = hardness in epm of the hardness standard solution.

V_1 = volume in mL of the hardness standard (100 mL) sample.

N_2 = concentration in epm of EDTA titrating solution.

V_2 = volume in mL of EDTA titrating solution required to reach the sky blue end point.

4.2.4 Class C, hardness indicator powder.

4.2.4.1 Hardness indicator, EBT, or equal. The procedure for calculating the percent of hardness indicator, EBT, or equal, shall be as follows:

a. Weigh 2.000 grams of the hardness indicator to be tested to the nearest 0.001 gram. Record the weight.

b. Transfer the sample to a 1-liter volumetric flask, dilute to the mark with reagent water, and mix thoroughly.

c. Weigh 0.100±0.001 gram of EBT. Record the weight.

d. Transfer the EBT to a 1-liter volumetric flask, and dilute to the mark with reagent water and mix thoroughly.

e. Pipet and transfer 4 mL, 6 mL, 8 mL, 10 mL, and 12 mL, respectively, of the EBT stock solution to 100-mL volumetric flasks.

f. Dilute to the mark with reagent water, and mix thoroughly.

g. Using 10-millimeter matched silica cells, measure the absorbances of the above standard solutions at a wavelength of 503 nanometers in accordance with ASTM E169.

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h. Determine the concentration of the EBT in the hardness indicator directly from the calibration curve by linear regression.

i. Calculate the percent EBT as follows:

$$Percent\ EBT = \frac{EBT,\ ppm \times 100}{Weight\ of\ sample \times 1000}$$

4.2.4.2 Sucrose. The sucrose shall be identified qualitatively by infrared spectroscopy (IR), using either internal reflection spectroscopy or alkali halide pressed pellet technique, in accordance with ASTM E1252. A finely ground mixture of hardness indicator and potassium bromide (KBr) shall be used to prepare a KBr pellet for an IR spectrum in accordance with ASTM E1252.

4.2.4.3 Hardness indicator performance. The test shall be performed using the hardness indicator being tested, hardness titrant meeting the requirements of class A of this specification, hardness buffer solution meeting class B of this specification, and hardness standard solution meeting class E of DOD-R-23679. The procedure for calculating the hardness of the hardness standard shall be as follows:

a. Measure 100 mL of the hardness standard solution using a volumetric pipet, and transfer into either a white porcelain casserole or into a clear Erlenmeyer placed on a white background.

b. Add 2 mL of buffer solution and 1 dipper of hardness indicator to a casserole.

c. Measure the beginning pH according to the procedures in ASTM D1293. Titrate to the endpoint, and measure the pH final endpoint.

d. Calculate the hardness of the hardness standard using the following equation:

$$H_1 = \frac{V_2 N_2}{V_1}$$

Where:

H_1 = hardness in epm of the hardness standard solution.

V_1 = volume in mL of the hardness standard (100 mL) sample.

N_2 = concentration in epm of EDTA titrating solution.

V_2 = volume in mL of EDTA titrating solution required to reach the sky blue end point.

4.2.4.4 Appearance. Visually inspect the hardness indicator powder.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The reagent chemicals covered by this specification are intended to be used in testing shipboard boiler feedwater in accordance with the procedure requirements of Naval Ships' Technical Manual S9086-GX-STM-020/220, Boiler Water/Feedwater Test and Treatment. These chemicals are used in conjunction with MIL-W-16642 and DOD-R-23679. Current stock of the reagent chemicals may be used until depleted; future requisitions for the reagent chemicals should refer to the current version of this specification.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class or classes of chemicals required (see 1.2).

c. Requirement for prohibited materials verification (see 3.1.1 and 4.1.1).

d. Test samples required (see 4.1.3).

e. Requirements for delayed delivery of the material pending results of tests (see 4.1.5).

f. When test reports are required.

g. Packaging and marking requirements (see 5.1, 6.3, and 6.4).

h. Safety data sheet (SDS), when required (see 6.5).

6.3 Suggested marking. In addition to any special marking specified (see 6.2), unit, interior, and exterior containers should be marked in accordance with MIL-STD-129 and should include the following:

a. National stock number.

b. Name and class of item.

c. Manufacturer's or contractor's name, or both.

d. Contract number.

e. Quantity/net content.

f. Date of manufacture, month, and year.

g. Expiration date.

h. "Note: Mandatory shelf-life material. Discard material after expiration date."

i. Lot or batch number.

6.4 Hazardous chemicals. As specified (see 6.2), each package unit of hazardous chemicals should have affixed thereto warning labels in accordance with Code of Federal Regulations (CFR) 49 CFR 171-199, 29 CFR 1910.1200, or appropriate Department of Defense instructions.

6.5 SDSs. When required (see 6.2), contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract.

6.6 Shelf-life. This specification covers an item where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings in accordance with MIL-STD-129 or FED-STD-123, preparation of materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at the time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, DoD Shelf-life Management Program. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil.

6.7 National stock numbers (NSN). Table V provides a list of NSNs assigned which correspond to this specification. This list may not include all possible NSNs associated with this document.

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TABLE V. List of applicable NSNs.

Class	Item Name	NSN	SLC [1/]	Unit of Issue
Class A	Hardness titrating solution	6810-01-072-1978	K [2/]	GL
Class B	Hardness buffer solution	6810-01-664-2218	M [3/]	BT (120 mL)
Class C	Hardness indicator powder	6810-01-390-4339	Q [4/]	BT (100 mL)

NOTES:

[1/] Shelf life codes

[2/] K – 18 Months

[3/] M – 24 Months

[4/] Q – 36 Months

6.8 Subject term (key word) listing.

Ammonium hydroxide

Buffer

Hardness in water

Hydrochloric acid

Indicator

Sodium hydroxide

Titrant

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
DLA – GS

Preparing activity:
Navy – SH
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